Lazy & IO

Season: 6 · FP Foundations · Category: Functional · Level: Intermediate

Intent

- Defer effects; run at the edge.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- A recipe card vs cooking the meal.

When to Use

- Pure functional style in Java 25 + Zazr.
- The design force named in Intent shows up more than once.

When Not to Use

- Throwing exceptions everywhere is 'fine' for a script.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.io;

import java.util.function.Function;
import java.util.function.Supplier;

/** Tiny IO effect — suspend side effects, interpret at the edge. */
public final class IO<A> {
    private final Supplier<A> thunk;

    private IO(Supplier<A> thunk) { this.thunk = thunk; }

    public static <A> IO<A> delay(Supplier<A> thunk) { return new IO<>(thunk); }
    public static <A> IO<A> pure(A a) { return new IO<>(() -> a); }

    public <B> IO<B> map(Function<? super A, ? extends B> f) {
        return new IO<>(() -> f.apply(thunk.get()));
    }

    public <B> IO<B> flatMap(Function<? super A, IO<B>> f) {
        return new IO<>(() -> f.apply(thunk.get()).unsafeRun());
    }

    /** Unsafe: runs effects. Call only at the program edge. */
    public A unsafeRun() { return thunk.get(); }
}
```

Modern Java 25 / Functional

- Primary examples use Zazr; VAVR shown in comparison episode.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/lazy/ + functional/io/
```

Pitfalls

- Zazr rejects null inside Option/collections.
- Keep interpreters at the edge.

Interview Q&A

- Q: Interview: what problem does Lazy & IO solve? A: Defer effects; run at the edge.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Defer effects; run at the edge.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: A recipe card vs cooking the meal. [0:40] Intent — State the design force in one sentence: Defer eff...